

Basic Functions

```
root@TCS1INMLJZJJJS3: ~  
root@TCS1INMLJZJJJS3:~# cat basic-function.sh  
#!/bin/bash  
read -p "enter name : " name  
greet() {  
    echo "hello $1"  
}  
greet "$name"  
  
add() {  
    read -p "enter num1 : " num1  
    read -p "enter num2 : " num2  
  
    echo "addition of number is $((num1+num2))"  
}  
  
add  
root@TCS1INMLJZJJJS3:~# ./basic-function.sh  
enter name : Phalguni  
hello Phalguni  
enter num1 : 10  
enter num2 : 20  
addition of number is 30  
root@TCS1INMLJZJJJS3:~#
```

Functions with Return Values

```
root@TCS1INMLJZJJJS3: ~  
root@TCS1INMLJZJJJS3:~# cat disk-check.sh  
#!/bin/bash  
  
check_disk() {  
    echo "----Disk Usage----"  
    df -h  
}  
  
check_memory() {  
    echo "-----Memory Usage-----"  
    free -h  
}  
  
check_disk  
check_memory  
root@TCS1INMLJZJJJS3:~# ./disk-check.sh  
----Disk Usage----  
Filesystem      Size  Used Avail Use% Mounted on  
none            3.9G   0  3.9G   0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2  
none            3.9G  4.0K  3.9G   1% /mnt/wsl  
drivers          475G  335G  141G  71% /usr/lib/wsl/drivers  
/dev/sdd         251G   5.1G  234G   3% /  
none            3.9G   80K  3.9G   1% /mnt/wslg  
none            3.9G   0  3.9G   0% /usr/lib/wsl/lib  
rootfs          3.9G  2.7M  3.9G   1% /init  
none            3.9G  516K  3.9G   1% /run  
none            3.9G   0  3.9G   0% /run/lock  
none            3.9G   0  3.9G   0% /run/shm  
none            3.9G   76K  3.9G   1% /mnt/wslg/versions.txt  
none            3.9G   76K  3.9G   1% /mnt/wslg/doc  
C:\              475G  335G  141G  71% /mnt/c  
tmpfs            781M   8.0K  781M   1% /run/user/0
```

Strict Mode — set -euo pipefail

```
root@TCS1INMLJZJJJS3: ~ × + v
root@TCS1INMLJZJJJS3:~# cat strict-mode.sh
#!/bin/bash
set -euo pipefail

echo "Testing undefined variable"
echo "hello $name"                                #name is undefined variable
echo "Testing failing command"
netstat                                           #This command will fail
echo "Testing failing in pipe"
false | true                                     #This would succeed without pipefail

root@TCS1INMLJZJJJS3:~# ./strict-mode.sh
Testing undefined variable
./strict-mode.sh: line 5: name: unbound variable
root@TCS1INMLJZJJJS3:~# |
```

Local Variables

```
root@TCS1INMLJZJJJS3: ~ × + v
root@TCS1INMLJZJJJS3:~# cat local-demo.sh
#!/bin/bash

local-function() {
    local name="Phalguni"
}

global-function() {
    city="Pune"
}

local-function
echo "name is $name"
global-function
echo "city is $city"
root@TCS1INMLJZJJJS3:~# ./local-demo.sh
name is
city is Pune
root@TCS1INMLJZJJJS3:~# |
```

Build a Script — System Info Reporter

```

root@TCS1INMLJZJJJS3: ~
#!/bin/bash
set -euo pipefail
system_info(){
    echo "System Information"
    hostnamectl | head -10 }

uptime_info(){
    echo "----System Uptime Information----"
    uptime }

disk_usage(){
    echo "----Disk Usage----"
    df -h }

memory_usage(){
    echo "----Memory Usage----"
    free -h }

cpu_usage(){
    echo "----CPU Usage----"
    ps aux --sort=-%cpu |head -6 }

main(){
    system_info
    uptime_info
    disk_usage
    memory_usage
    cpu_usage
}
main

```

```

root@TCS1INMLJZJJJS3: ~
root@TCS1INMLJZJJJS3:~# ./system_info.sh
System Information
Static hostname: TCS1INMLJZJJJS3
Icon name: computer-container
Chassis: container
Machine ID: 281366fdd28b40d5a3f73fae9bd2fa6a
Boot ID: 28af3aa2b2cd4a8a98a2eaaaf3297505e
Virtualization: wsl
Operating System: Ubuntu 22.04.5 LTS
Kernel: Linux 6.6.87.2-microsoft-standard-WSL2
Architecture: x86_64
----System Uptime Information----
16:45:33 up 1:10, 1 user, load average: 0.00, 0.00, 0.00
----Disk Usage----
Filesystem      Size  Used Avail Use% Mounted on
none            3.9G  0  3.9G  0% /usr/lib/modules/6.6.87.2-microsoft-standard-WSL2
none            3.9G  4.0K  3.9G  1% /mnt/wsl
drivers         475G  335G  141G  71% /usr/lib/wsl/drivers
/dev/sdd        251G  5.1G  234G  3% /
none            3.9G  92K  3.9G  1% /mnt/wslg
none            3.9G  0  3.9G  0% /usr/lib/wsl/lib
rootfs          3.9G  2.7M  3.9G  1% /init
none            3.9G  520K  3.9G  1% /run
none            3.9G  0  3.9G  0% /run/lock
none            3.9G  0  3.9G  0% /run/shm
none            3.9G  76K  3.9G  1% /mnt/wslg/versions.txt
none            3.9G  76K  3.9G  1% /mnt/wslg/doc
C:\             475G  335G  141G  71% /mnt/c
tmpfs           781M  8.0K  781M  1% /run/user/0
----Memory Usage----
              total        used        free      shared  buff/cache   available
Mem:          7.6Gi        373Mi        7.1Gi         3.0Mi        182Mi        7.1Gi
Swap:         2.0Gi           0B         2.0Gi
----CPU Usage----
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root      1142  0.4  0.0  14696  6912 ?        Ss   16:45   0:00 /lib/systemd/systemd-hostnamed
root         1  0.1  0.1 165968 10988 ?        Ss   15:35   0:06 /sbin/init
root         2  0.0  0.0   3120  1920 ?        Sl   15:35   0:00 /init
root         6  0.0  0.0   3136  1792 ?        Sl   15:35   0:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --log-truncate
root        87  0.0  0.1  31428 10880 ?        S<s  15:35   0:00 /lib/systemd/systemd-journald

```